

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
II SEMESTER 2013-2014
EEE/CS/INSTR F241 MICROPROCESSOR PROGRAMMING AND INTERFACING
COMPREHENSIVE EXAMINATION-A (OPEN BOOK)
13-05-2014

MARKS: 30

ID No:

Name: **SOLUTIONS**

Sec:

DURATION: 60 MIN

Note: This in-built question/answer paper contains fifteen questions, each carrying 2 marks. Use last few pages of answer sheet of PART-B to do your rough work. You can submit PART-A anytime and collect PART-B question paper.

Recheck Request:

Q1. The expression $N_{\text{a. of clocks}} = 22(CX-1) + 1,441,802 (DX-1) + 36$ introduces a delay of 0.758 sec for $CX=4150H$ and $DX=0006H$ in an x86 CPU based program. The frequency of operation of CPU is 10 MHz. Compute the minimum and maximum delay which could possibly be achieved using the expression.

Minimum delay: when $CX=DX=1 \rightarrow 3.6\mu s$

Maximum delay: when $CX=DX=65535 \rightarrow 9449 \text{ sec}$

Q2. A MACRO is written by a user to push the contents of EAX, EBX, ECX and EDX registers in order onto the stack. Show the contents using register names and their physical addresses in the stack, after the code associated with the MACRO is executed. Initially, assume $SS=0000H$ and $SP=1000H$.

$SP=1000h \rightarrow ?$

$SP=0FFCh \rightarrow EAX$

$SP=0FF8h \rightarrow EBX$

$SP=0FF4h \rightarrow ECX$

$SP=0FF0h \rightarrow EDX$

Q3. If a media has 256 bytes/sector and the maximum number of root directory entries specified in BIOS Parameter Block is 612, then total number of sectors reserved for root directory is:
 $32 \text{ bytes/entry} \times 612 \text{ entries} / 256 \text{ bytes/sector} = 77 \text{ sectors}$

Q4. If the 8 byte descriptor of a segment in 80286 is 00 00 FF 32 00 00 00 FF, then the size and the starting address of the segment is

Size = 256 bytes

Starting Address: 320000H

Q5. If $AX=FFFFH$ and $CL=02H$. What is the result of executing the instruction $DIV CL$?

Results in divide by 0 error.

Q6. The USB protocol uses NRZI encoding scheme to encode the data before transmitting it. Use this scheme to encode the following bits: 111101101
111100011

Q7. Out of the three inputs of 8086, NMI, INTR and HOLD, which input has the highest priority?

HOLD input

Q8. Compute the time taken to transmit three ASCII characters via 16550 operating at 19200 baud rate with 8 bit data, odd parity and one stop bit.

Bit duration = $1/19200 \text{ bps} = 52.083 \mu\text{s}$.

Each character has 1 start (mark) bit, 8 character bits 1 parity bit and 1 stop bits = 11 bits

The whole transmission = $(3 \text{ char} * 11 \text{ bits/char}) / 19200 = 1.72 \text{ ms}$

Q9. For 80486 CPU, what are the four different methods using which an application running at the lowest privilege level can access the procedure running at the highest privilege level?

CALL gate

Interrupt gate

Trap gate

Task gate

Q10. For 80386, answer True/False for the following statements:

- The CPL of the currently executing procedure is normally equal to the privilege level of the code segment from which the instructions are being fetched. **True**
- A CALL gate with a DPL of 2 can be used by a program with a CPL of 2 to access a system service at level 0 but the same gate cannot be used by a program with a CPL of 3. **True**

Q11. (a) A cache is divided into 16 sets of 4 lines each. The main memory contains 4K blocks of 128 words each. The format of the main memory addresses would be

TAG:	SET:	WORD/OFFSET:
8	4	128

(b) Conflict miss is higher in direct mapped cache as compared to set associative cache (True/False): **True**

Q12. A computer has a cache, main memory and a hard disk. If an operand is in the cache, 20 ns are required to access it. If it is in the main memory but not in cache, 60 ns are needed to load it into cache and then it is accessed by CPU. If the operand is not in main memory, 12 ms are required to fetch it from disk, followed by 60 ns to copy it into the cache and then the operand is accessed by the CPU. The cache hit ratio is 0.9 and the main memory hit ratio is 0.6. What is the average time in nano seconds required to access the required operand on this system?

Average time = $[20 \times 0.9] + [(60 + 20) \times (0.6 \times 0.1)] + [(12000000 + 60 + 20) \times (0.1 \times 0.4)] = 480026 \text{ ns}$

Q13. The Intel 8088 consists of a bus interface unit (BIU) and an execution unit (EU), which forms a 2-stage pipeline. The BIU fetches instructions into a 4-byte instruction queue.

- a) Suppose the tasks performed by the BIU and EU take about equal time. By what factor does pipelining improve the performance of the 8088? Ignore the effect of branch instructions.

Factor of 2

- b) Repeat the calculation assuming that the EU takes twice as long as the BIU.

Factor of 1.5

Q14. Consider a system employing interrupt-driven I/O for a particular device that transfers data at an average of 8 KB/s on a continuous basis. Assume that interrupt processing takes about $100\mu\text{s}$ (i.e., the time to jump to the interrupt service routine (ISR), execute it, and return to the main program). Determine what fraction of processor time is consumed by this I/O device if it interrupts for every byte.

The device generates 8000 interrupts per second or a rate of one every $125\mu\text{s}$. If each interrupt consumes $100\mu\text{s}$, then the fraction of processor time consumed is $100/125 = 0.8$

Q15. A pipelined processor has a clock rate of 2.5 GHz and executes a program with 1.5 million instructions. The pipeline has five stages, and instructions are issued at a rate of one per clock cycle. What is the throughput (in MIPS) of the pipelined processor?

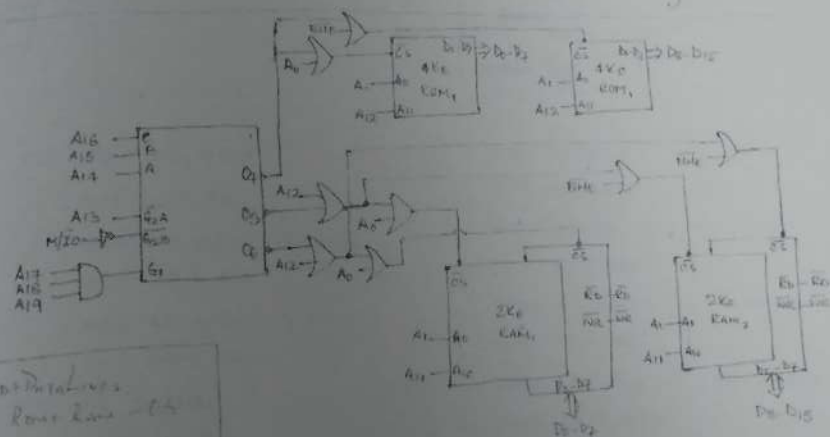
2500 MIPS as each instruction is issued at a rate of one per clock cycle.

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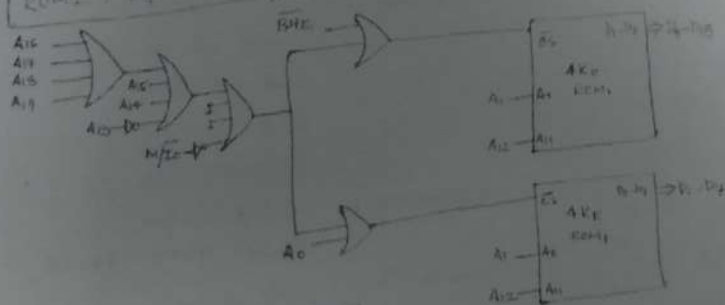
Qn 1

A ₁₆	A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	1
1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1

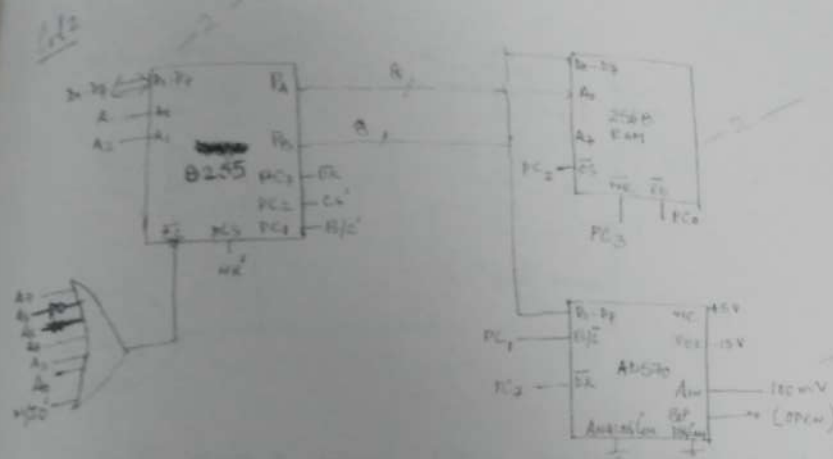
A₁₆
 0 } 4K RAM₁
 1
 0 } 4K RAM₂
 1
 0 } 8K ROM₁
 1
 0 } 8K ROM₂
 1



Add + Data Lines
 CF ROM₁ RAM₁ → C4
 RAM₁ + RAM₂ → C4
 ROM₁ → C4
 ROM₂ → C4



NOTE: MUST BE
 ADDRESS DECODING



```

X2: MOV AL, 8AH
   OUT 46H, AL
   MOV AL, 05H ; EC DIRECTION
   OUT 43H, AL
   MOV AL, 02H ; SEC
   OUT 44H, AL
X1: IN AL, 44H
   AND AL, 80H
   JNZ X1

```

NOP ; 2000 DELAY

IN AL, 42H

MOV BL, AL

MOV AL, 03H

OUT 44H, AL

MOV CX, 02 ; 2000 DELAY

DELAY: LOOP DELAY

MOV AL, 04H ; EC DIRECTION

MOV AL, FFH ; TOP MEMORY

OUT 40H, AL ; LOCATION

MOV AL, 85H ; P0 o/p

OUT 46H, AL

MOV AL, BL

OUT 43H, AL

MOV AL, 06H ; NR ENABLE

OUT 44H, AL

MOV AL, 05H ; EC DIRECTION

OUT 44H, AL

JMP X2

Get Data Mode - 01H

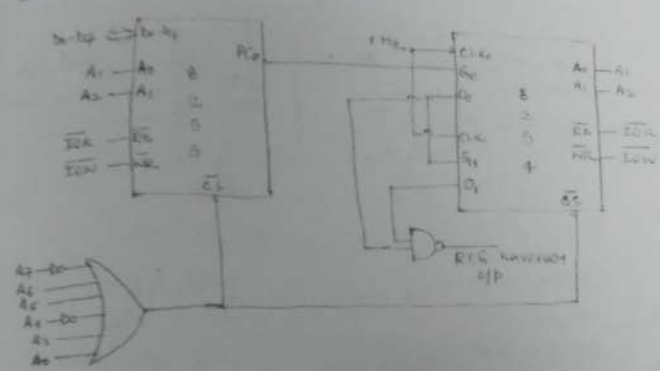
EN/Pullup - 01H

Delay Insertion - 03H

Read AHC & Write

Memory - 04H

Ex. 3



If you apply a signal (1 Hz in this case) which is not derived from the system clock to CLK input of 8254 in mode 2 or 3, then the gate input of counter must be pulsed low just after the count is written to counter and then high to enable the counter. So, 8255 has to be used.

Counter 0 in mode 2 with count value of 6.
Counter 1 in mode 1 with count value of 3.

INIT PROC NEAR

```
MOV AL, 00D1D1D1
OUT 96H, AL
MOV AL, 6
OUT 90H, AL
MOV AL, 80H
OUT 96H, AL
MOV AL, 00H
OUT 94H, AL
MOV AL, D1H
OUT 94H, AL
```

Initialize
CF

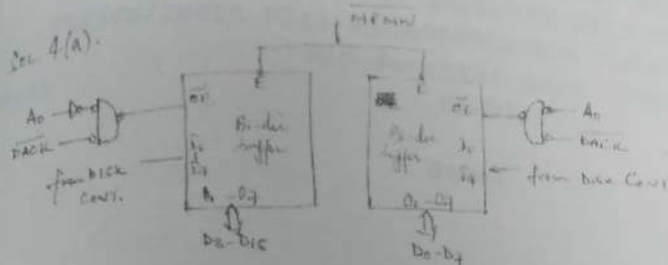
Initialize
BSS

```
MOV AL, 01010011
OUT 96H, AL
MOV AL, 3
OUT 92H, AL
RET
ENDP
```

Initialize
C2

8255 - 05 M
ALP - 03 M
DESBING LOWC - 01 M
Final off - 06 M
(WITH CURRENT MODES)

Sec 4(a).



4(b) model tiny
data

CH0 equ F0H
CHOC equ F1H
CH1 equ F2H
CHIC equ F3H
CR equ FBH
ERB equ F9H
MR equ FBH
CFF equ FCH
MASKS equ FFH
STATUS equ FBH

Code

Startup

MOV AX, 5000H
MOV BX, AX
MOV SI, 0000H
MOV FS, AX
MOV DS, 1000H

CHL AX, 4
ADD AX, SI
OUT CH0, AL
MOV AL, AH
OUT CH0, AL

MOV AX, ES
CHL AX, 4
ADD AX, DI
OUT CH1, AL
MOV AL, AH
OUT CH1, AL

MOV CX, 4096
MOV AX, CX
DEC AX
OUT CHIC, AL
MOV AL, AH
OUT CHIC, AL

MOV AL, 10010000b
OUT MR, AL
MOV AL, 10000101b
OUT MR, AL

MOV AL, 10100001b
OUT CR, AL

MOV AL, 0CH
OUT MASKS, AL

MOV AL, 4
OUT REG, AL

XI: IN AL, STATUS
CMP AL, 0FH
JNZ XI

EXIT
END